SPARK Challenge 2025/26

Integrated Smart Monitoring Unit for Adult Dengue Patient Care in Sri Lankan Hospitals

Nexora

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1) Abstract

Dengue fever continues to place a significant burden on the Sri Lankan healthcare system, particularly during seasonal outbreaks where hospitals face overcrowding and limited clinical resources. Adult dengue patients require close and continuous monitoring, especially during the transition from the febrile phase to the critical phase, where rapid deterioration can occur within hours. At present, this monitoring is largely manual, relying on frequent nursing observations and paper-based charts, which increases the risk of delayed detection of shock and plasma leakage. This project proposes a Raspberry Pi–based integrated monitoring unit designed specifically for adult dengue patient management in Sri Lankan hospitals. The system continuously monitors key physiological and clinical parameters, analyzes trends based on national dengue management guidelines, and provides timely alerts to healthcare staff. By improving early detection, reducing staff workload, and standardizing monitoring practices, the proposed solution aims to reduce preventable complications and deaths due to dengue.

2) Problem Statement

Dengue affects thousands of adults in Sri Lanka annually, with a substantial proportion requiring hospital admission for close monitoring. The most dangerous period of dengue illness is the critical phase, typically lasting 24–48 hours, during which patients are at risk of plasma leakage, shock, bleeding, and organ failure. Current hospital practices require frequent measurement of vital signs, urine output, hematocrit (PCV), and fluid balance, often at hourly intervals during the critical phase. In busy wards with high nurse-to-patient ratios, maintaining this level of vigilance is challenging, increasing the likelihood of missed early warning signs. A nurse caring for multiple dengue patients may need to manually document dozens of readings per shift, leaving limited time for trend analysis and early intervention. This creates an opportunity for an integrated technological solution that supports clinical decision-making while fitting seamlessly into existing hospital workflows.

3) Background and Motivation

Dengue fever remains a major public health challenge in Sri Lanka, with increasing incidence and severity influenced by climate change–related factors such as rising temperatures, irregular rainfall patterns, and extended mosquito breeding seasons. These climate-driven conditions place additional strain on healthcare systems, particularly during outbreak periods, where hospitals must manage a high volume of dengue patients requiring continuous and accurate monitoring of multiple vital and laboratory parameters. At present, dengue patient monitoring in most Sri Lankan hospitals—both government and private—is largely dependent on manual procedures. These include manual measurement of vital signs, handwritten observations, and manual data entry into records or systems, followed by repeated manual checking by nurses and doctors. This process is time-consuming, increases the risk of human error, and significantly adds to the workload of healthcare staff, especially nurses who are responsible for frequent monitoring in dengue wards. The idea for this project originated from discussions with an intern doctor, who highlighted that manual monitoring and data recording have been persistent challenges for hospital staff over many years. Based on this insight, our team conducted an anonymous survey among doctors and nurses to assess whether this issue represents a real and unresolved problem in clinical practice. The survey findings confirmed that manual data entry and manual checking remain major pain points for hospital staff and that existing solutions do not adequately address these challenges. Due to privacy and ethical considerations, no stakeholder-identifiable information is disclosed; however, a summary of the anonymous survey results is attached to this document as supporting evidence. Previous attempts to address dengue monitoring have focused on isolated or parameter-specific solutions, such as the development of devices to monitor CRFT alone. While such approaches demonstrate technical potential, they are limited in scope and fail to solve the core issue identified by stakeholders: the lack of an integrated, automated, multi-parameter monitoring and data management system, and the continued dependence on manual data entry. As emphasized by hospital staff during the survey, eliminating or minimizing manual data recording is a critical requirement that remains unmet. Motivated by direct clinical insights, validated through survey data, and guided by the need for a scalable and context-appropriate solution, our team aims to develop a smart dengue patient monitoring system that reduces workload, improves data accuracy, and enhances clinical responsiveness. The project is designed with the operational realities of Sri Lankan hospitals in mind and seeks to strengthen healthcare resilience in the face of climate-sensitive disease outbreaks.

4)Solution

The proposed Smart Dengue Patient Monitoring System is an integrated solution that combines auto mated vital parameter monitoring with a centralized digital platform for data storage, visualization, and analysis. The system is intended for use by doctors and nurses in government hospitals, private hospitals, and specialized dengue wards, with the primary goal of reducing manual workload while improving the accuracy and timeliness of patient monitoring.

Hardware-Based Automated Monitoring System

A Raspberry Pi–based hardware platform is used to automate the monitoring of selected vital and clinical parameters. These include body temperature, pulse rate and pulse volume, respiratory rate, blood pressure and pulse pressure, CRFT, PCV, platelet count, WBC, and IP/OP status. The specific parameters monitored can be configured based on stakeholder requirements and clinical needs. Data acquisition is handled automatically through a programmed system, incorporating sensor-based techniques and advanced approaches such as computer vision where applicable. This automation significantly reduces the need for manual measurements and manual data entry.

Web-Based Application and Data Management Platform

The second component is a secure web application designed to store, record, and manage patient monitoring data in real time. Data collected by the hardware system are transmitted via Wi-Fi and displayed on a web dashboard accessible to authorized doctors and nurses, as well as on a local display within the ward. The system maintains comprehensive patient histories and enables trend analysis, allowing medical staff to observe changes over time rather than relying solely on isolated readings. A real-time alert mechanism notifies healthcare staff when monitored parameters exceed predefined clinical thresholds, enabling early detection of warning signs and timely medical intervention.

5)Feasibility and Uniqueness

- Here is a comparison of some of the best existing solutions with our proposed system for addressing this problem.

Manual Paper Cards	• Very slow & manual → delays in surveillance & reporting. Paper forms take time to fill and process. • Errors & missing data common due to handwriting & transcription. • Hard to search or analyze data; no dashboards or real-time alerts.	• Automated data capture + database: real-time inputs, searchability & alerts. • Digitized storage: no risk of lost/damaged forms. • Faster decision making for clinical decisions & outbreaks	https://cdn.who.int/media/docs/default-source/searo/dengue/webinar-series-on-dengue-outbreak/day3/5.-ms-geethani-udugamakorala_adult-dengue-monitoring.pdf?sfvrsn=ac5b9970_1
DHIS2 (disease surveillance platform)	• Not designed for detailed clinical vitals or patient-level real-time measurements (possible but not clinical monitoring by default). • Can be complex to configure clinical workflows in a hospital ward • Not inherently capturing wearables/IoT sensor vitals (needs integration + custom work).	• Built-in IoT vitals integration (sensor → dashboard). • Patient-centric record + dengue workflows (platelets, vitals, alerts). • Customizable forward/field use vs standard weekly reporting.	https://dhis2.org/disease-surveillance/
OpenMRS (open-source EMR	• Designed for general EMR records, not optimized as a dengue monitoring + real-time data system. • Requires significant customization to handle vitals + alerts per disease protocol. (general EHR platform) • Not inherently integrated with IoT sensors without extra modules.	• Pre-built disease data + IoT + alerts for dengue severity indicators. • Simpler, disease-specific workflow focused on dengue care needs. • Real-time dashboard & notifications (nurse/doctor focused).	https://en.wikipedia.org/wiki/OpenMRS

Feasibility

1.Technological Feasibility

- Uses readily available microcontrollers (Raspberry Pi or equivalent) with medically safe sensors for key vitals.
- Manual entry fields ensure usability even without full lab automation.
- A secure, cloud-based database makes data easy to store, access, and back up, removing the risk of lost paper charts.
- The system includes a built-in analytics layer, enabling easy data analysis, trend visualization, and automatic detection of abnormal patterns.

2. Operational Feasibility

- Nurses currently record more than 12 parameters manually.

  Automating vital signs reduces workload so they can focus more on clinical care.

- The UI mirrors existing dengue monitoring charts, requiring minimal training and integrating naturally into hospital workflow.
- The digital system makes data easy to manage, retrieve, filter, and compare across hours, days, and phases (febrile → critical).
- Automatic alert rules for dengue-specific danger signs (low pulse pressure, rising PCV, CRFT >2s, high pulse rate, etc.) improve patient safety.

3. Cost Feasibility

- Partial automation keeps the system budget-friendly, especially for government hospitals.
- Reduces the cost of maintaining large volumes of paper files.
- Saves time and labor costs associated with repetitive manual measurements and rewriting charts.
- Scalable architecture allows hospitals to add more automation modules later without large investments.

Uniqueness

1. Hybrid Automation Designed Exactly for Dengue Workflow

Our system is uniquely built to match the official dengue monitoring structure
It offers:

- Automated vitals
- Manual lab integrations
- Phase-based charting (Febrile & Critical)
  This ensures a perfect fit with existing hospital practice.

2. First-of-its-kind Digital Dengue Monitoring + Automated Alerts

The system actively monitors danger signs such as:

- Pulse pressure < 25mmHg
- CRFT > 2 seconds
- Rising PCV trends
- Sudden pulse rate changes
  This is not possible with current paper-based charts.

3. Easy Data Analysis, Storage & Management

Unlike manual systems, our platform provides:

- Clean digital records stored securely in the cloud
- Trend graphs for vitals, PCV, urine output, fluid balance
- Automatic identification of critical-phase entry
- Hospital-level and national-level dashboards
- Ability to export data for research, training, and audits
  This turns raw clinical data into actionable insights.

4. Scalable Future Ready Architecture

Our design allows seamless upgrades, such as:

- Lab-machine integration for automated PCV/platelet uploads
- AI-based prediction of shock risk
- Expansion to pediatric dengue monitoring
- Integration across multiple hospitals for a national dengue surveillance dashboard
  This scalability makes the system competitive, future-proof, and suitable for long-term impact.

5. Affordability + Practicality

While international systems focus on expensive full automation, our model is purpose built for Sri Lanka affordable, realistic, and immediately usable, making adoption practical even in overcrowded wards.

6)SDG Alignment

Strengthening Climate-Resilient Healthcare
• Enhances hospital response during climate-driven dengue outbreaks
• Supports early detection of critical cases
• Builds long-term, data-driven healthcare resilience

Reducing Paper Use and Carbon Footprint
Paper Usage Estimation
• 4pages/day × 5 days = 20 pages per patient
• 100 patients ⇒ 2,000 A4 sheets per season
• 1A4sheet ≈ 5 gCO_2

• Total emissions ≈ 10 kg CO_2 per ward per season

With the proposed digital system, paper usage is nearly zero, achieving approximately 100% paper reduction.

Efficient Resource Management

• Improved allocation of beds, fluids, and staff
• Reduced errors in manual fluid balance charts
• Better forecasting of patient load using analytics

Human Workload Reduction

• Manual workload: ≈ 67 nurse-hours per season
• Automated system reduces workload by 70–80%
• Estimated savings: 45–55 nurse-hours per ward per season

Digital Innovation for Climate Adaptation

The use of IoT sensors, cloud infrastructure, and real-time analytics demonstrates how digital health systems can support adaptation to climate-driven health challenges, aligning with SDG 13.1.

Supporting Public Health Policy

• Enables national dengue surveillance
• Supports early outbreak warning systems
• Facilitates evidence-based policymaking

7)Marketing and Business Strategy

The Ministry of Health spends millions annually on dengue management. We are proposing a smart dengue patient monitoring system to monitor vital parameters with ease. Our marketing strategy goes like this. It is focusing on the transition from reactive care to proactive monitoring.

Phase 1: Clinical Validation

- At this stage, we will deploy our prototype at selected general hospitals and the IDH. During this time nurses will perform their usual medical tests while our system simultaneously records the same parameters. The resulting “validation report” signed by an IDH consultant acts as our primary sales tool. We will develop our system with full integrity to perform at maximum efficiency and accuracy.

Phase 2: Regulatory & Certification

- In this phase we will move from an experimental project to a legally marketable medical device. We will submit our design documentation, clinical data from Phase 1 and safety test results to the National Medicines Regulatory Authority (NMRA) of Sri Lanka. Our goal is to obtain the “Seal of Safety”. This will certify our device is electrically safe for hospitals, and our web panel handles patient data securely.

Phase 3: Launch

- Once we have the data and the certification, we will move to the national stage. This stage is about positioning our device as an essential part of Sri Lanka’s healthcare infrastructure.

8)Timeline

Phase 1: Research & Problem Identification (October 2025 – September 2025)

- Problem identification: identified the necessity for a smart dengue monitoring system.
- Problem validation: Conducting discussions with hospital staff and running the anonymous survey.
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Phase 2: System Design & Proposal Submission (November 2025 – December 2025)

- Compiling research data and Proposal submission
- Finalizing the selection of sensors (Pulse, BP, Temp) and designing the Raspberry Pi communication protocol.

Phase 3: Hardware Acquisition & Core Development (January 2026 – February 2026)

- Receiving the Raspberry Pi kit and interfacing medical sensors.
- Developing the web-based data acquisition system and setting up the local database for vital signs storage.
- Creating the initial breadboard prototype to test sensor accuracy.

Phase 4: Integration & MVP Testing (March 2026 – April 2026)

- Developing the web dashboard and coding the alert algorithms
- 3D printing a robust, hospital-safe casing for the main unit.
- Completion of the Minimum Viable Product (MVP) and submission of the demonstration video

Phase 5: Clinical Pilot & Validation (May 2026 – June 2026)

- Deploying the unit at the IDH or a General Hospital for the "Clinical Validation" pilot mentioned in the marketing strategy.
- Running the system parallel to manual nursing charts to generate the "Validation Report."

Phase 6: Regulatory Preparation & Final Pitch (July 2026)

- Compiling safety test results and clinical data for future NMRA submission.
- Finalizing the cost structure and pitch deck.

stakeholder feedbacks:
link: -
https://docs.google.com/forms/d/11YFUzs0B9vlMuRCZ5Ko30mNIhCJuUeoQPEj0lYKmEa8/edit?ts=69418505#responses